

~~ AquaVeda ~~

Watershed Impact Assessment Report

Hiware Bazar

District: Ahmednagar | State: Maharashtra

Analysis Period: 2019 - 2024

Health Score: 82/100 Excellent

Generated by AquaVeda - Satellite-Based Watershed Monitor

Data Source: Sentinel-2 (ESA/Copernicus) via Google Earth Engine

Report Date: August 23, 2026

Executive Summary

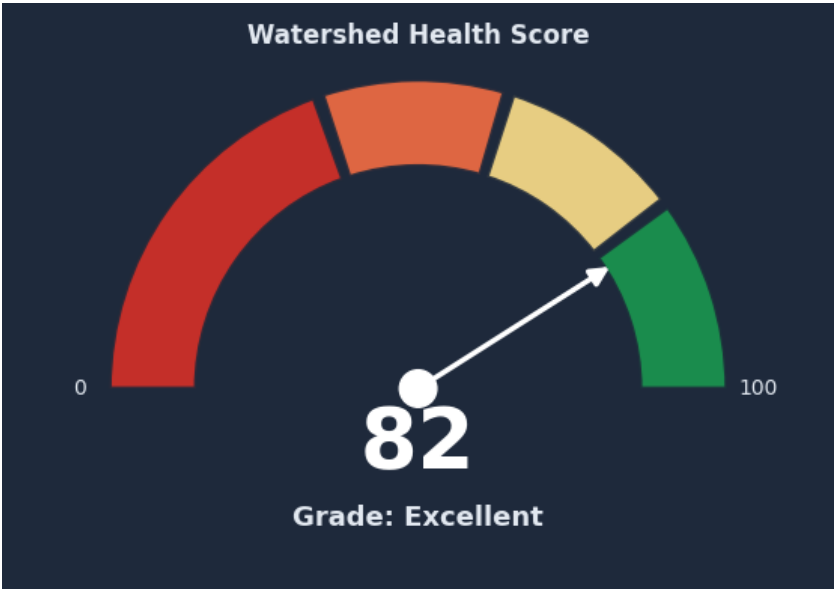
Overall Health Score: 82/100 - Grade: Excellent

Key Findings:

- * Vegetation cover improved by 0.30 NDVI units (NDVI: 0.28 -> 0.58)
- * Water body area grew from 3.2 ha to 12.8 ha (+300%)
- * High erosion risk area reduced by 66%
- * Analysis period: 2019 to 2024 (5 years of intervention)

Score Breakdown:

Component	Score	Max	Grade
Vegetation Health	30	35	A
Water Resources	32	35	A
Erosion Control	15	20	B
Sustainability	5	10	C
TOTAL	82	100	A



Vegetation Analysis

Metric	Before	After	Change
Mean NDVI	0.280	0.580	+0.300
Improved Area (ha)	-	520	61.2% of watershed
Declined Area (ha)	-	35	-

The NDVI increased by 0.30 units over the project period, representing a 107% relative improvement in vegetation health. Values above 0.4 indicate a healthy, productive landscape.

Water Resources Analysis

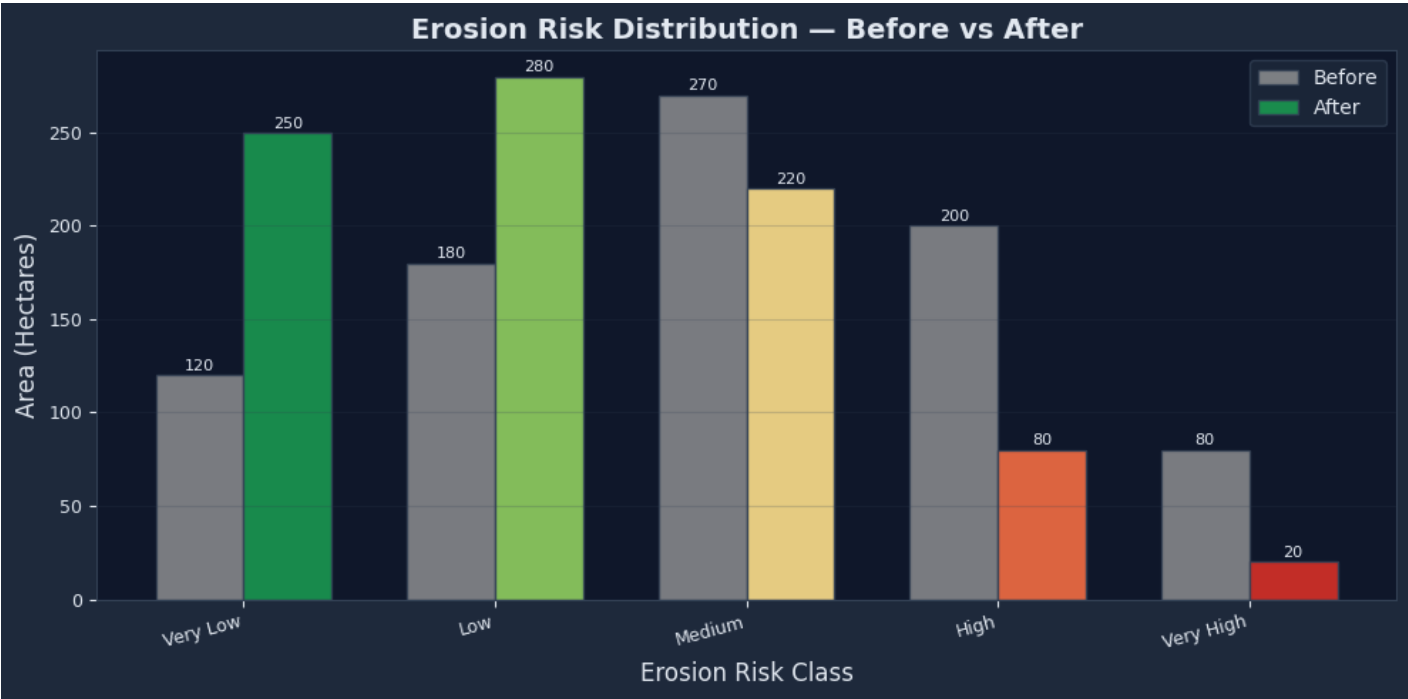
Metric	Before	After	Change
Water Body Area (ha)	3.2	12.8	+10 ha
Change Percent	-	-	+300%
Months with Water/yr	4	9	-

Water body coverage expanded from 3.2 ha to 12.8 ha - a 300% increase - demonstrating the effectiveness of check dams, farm ponds, and percolation structures in improving water retention.

Erosion & Land Use Change

Risk Class	Before (ha)	After (ha)	Change (ha)
Very Low	120	250	+130
Low	180	280	+100
Medium	270	220	-50
High	200	80	-120
Very High	80	20	-60

High and Very High erosion risk areas were reduced by 66%, reflecting improved ground cover and soil conservation measures. Contour trenches, gabion structures, and vegetative cover collectively contribute to this improvement.



Recommendations

Rec 1	Continue maintaining check dam #3 near stream junction
Rec 2	Plant fruit trees on recovered barren patches (NW zone)
Rec 3	Consider drip irrigation for newly cultivated areas
Rec 4	Install rain gauge at percolation tank #2 for monitoring

Field Verification Log

Total records: 4

Type	Status	Latitude	Longitude	Date
Check Dam	Functional	19.3520	74.5540	2024-06-15

Notes: Check dam on Hiware stream - holding water well after monsoon

Notes: Farm pond near Patil farm - used for supplemental irrigation

Notes: Contour trenches on hillslope - visible soil moisture improvement

Notes: Percolation tank - spillway damaged, needs maintenance

- Farm Pond
- Contour Trench
- Percolation Tank